



WEEK 2 PORTFOLIO PROJECT

Enterprise Infrastructure Planning for a Startup Company

ABC Startup Solutions

ITEP 414 – System Administration and
Maintenance

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Republic of the Philippines
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PART 1 — COMPANY PROFILE

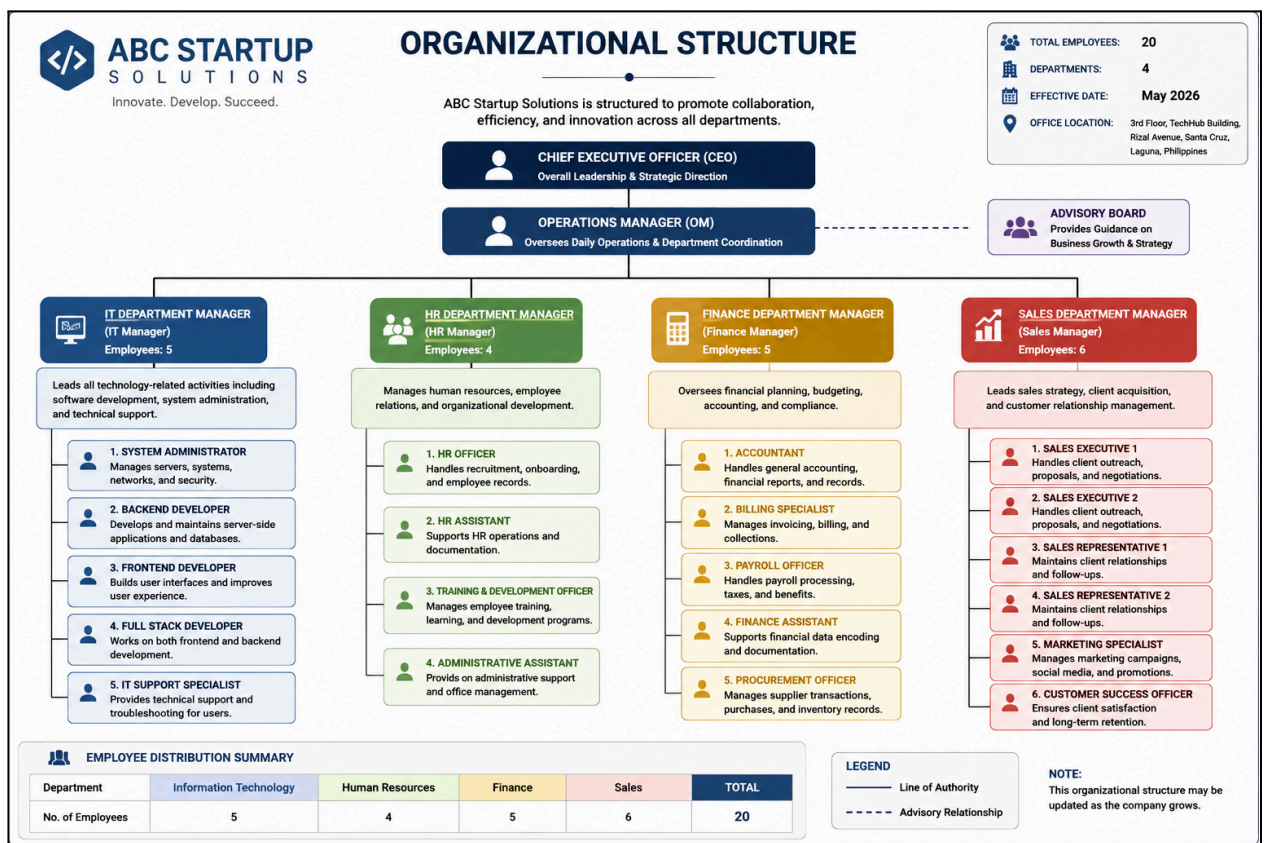
Company Name: ABC Startup Solutions

Nature of Business: A fictional software development startup providing custom web and mobile applications, business automation systems, technical consulting, and maintenance services for small and medium-sized organizations.

Vision: To become a trusted technology partner known for reliable, secure, and practical software solutions that help organizations work smarter.

Mission: To deliver maintainable software and responsive technical services through skilled people, secure infrastructure, continuous learning, and customer-focused support.

Office Location: 3rd Floor, Innovation Hub, Brgy. San Isidro, Santa Cruz, Laguna (fictional location for academic use).





Employee Distribution

Department	Employees	Primary IT Needs
Information Technology	5	Development workstations, server access, testing tools, network administration
Human Resources	4	Office productivity, HR records, secure printing and file access
Finance	5	Accounting/office applications, restricted files, secure printing and backups
Sales	6	CRM/browser work, presentations, email, Wi-Fi and shared resources
TOTAL	20	—

PART 2 — ENTERPRISE HARDWARE INVENTORY

The inventory allocates one primary endpoint per employee while giving IT staff mobile-capable devices for administration and development. Quantities are intentionally sized for 20 employees plus shared infrastructure.

Asset ID	Hardware	Qty	Department	Purpose
HW-001	Business Desktop PC	15	HR, Finance, Sales	Standard office productivity, browser, CRM, accounting and docu
HW-002	Business Laptop	5	IT	Development, administration, testing, meetings and mobility
HW-003	Rack/Tower Server	1	IT	Central services, authentication, file/application services and mon
HW-004	Enterprise Router	1	IT	Routing, NAT, VPN and network gateway functions
HW-005	Managed Gigabit/2.5G Switch	2	IT	Connect wired endpoints and provide capacity/redundancy
HW-006	Laser Multifunction Printer	2	Shared	HR/Finance and Sales/IT printing and scanning
HW-007	UPS 1500VA	6	IT / Shared	Power protection for server, network core, NAS and critical device
HW-008	Wireless Access Point	2	Office	Secure employee Wi-Fi and guest Wi-Fi
HW-009	NAS Storage	1	IT	Central backup repository and file storage
HW-010	External Backup Drive	2	IT	Offline backup rotation
HW-011	24-inch Monitor	10	IT / Finance / Sales	Additional screen space for development and productivity

Selection justification: 15 desktops cover the non-IT workforce; five laptops support IT mobility and administration. Two managed switches and two access points provide room for segmentation and future growth. The server, NAS, UPS units and offline drives protect critical services and business data.



Part 3 — Enterprise Software Inventory

Software	Version / baseline	License	Purpose
Windows 11 Pro	25H2	Commercial	Primary desktop OS with business security and management features
Ubuntu Server	24.04 LTS	Open source	Server platform for selected internal services, monitoring and dev
Microsoft 365 Apps	Current channel	Subscription	Word, Excel, PowerPoint, Outlook and collaboration/productivity
Visual Studio Code	Current stable	Free	Software development and configuration editing
Git	Current stable	Open source	Distributed source-code version control
GitHub Desktop	Current stable	Free	GUI workflow for Git repositories and collaboration
VirtualBox	7.x	Open source / free	Virtual machines for testing, labs and isolated environments
Google Chrome	Current stable	Free	Business web access and web application testing
Microsoft Defender	Windows-integrated	Included	Endpoint malware and threat protection
AnyDesk	Business/enterprise plan	Commercial	Authorized remote support and troubleshooting
7-Zip	Current stable	Open source	File compression, extraction and archive management

Version note: “Current stable” is used where the exact build changes frequently. Before procurement, the administrator should record the exact approved build and update cadence in the asset register.

Part 4 — Enterprise Network Inventory

Asset ID	Equipment	Qty	Specification / Baseline	Purpose
NET-001	ISP Modem/ONT	1	Provider-supplied business gateway	Terminates the internet service
NET-002	Router	1	Enterprise-class, VLAN/VPN capable	Gateway routing, NAT, VPN and policy enforcement
NET-003	Firewall	1	Next-generation firewall	Threat filtering, segmentation, logging and secure remote access
NET-004	Managed Switch	2	24-port managed, VLAN capable	Wired connectivity, VLAN segmentation and centralized
NET-005	Wireless Access Point	2	Wi-Fi 6/6E business AP	Employee and guest wireless connectivity
NET-006	Patch Panel	2	24-port Cat6	Structured cable termination and organization
NET-007	CAT6 Cable	1 lot	Approx. 600 m, certified Cat6	Horizontal cabling and patching
NET-008	RJ45 Connectors	100	Cat6 compatible	Cable termination and maintenance stock



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